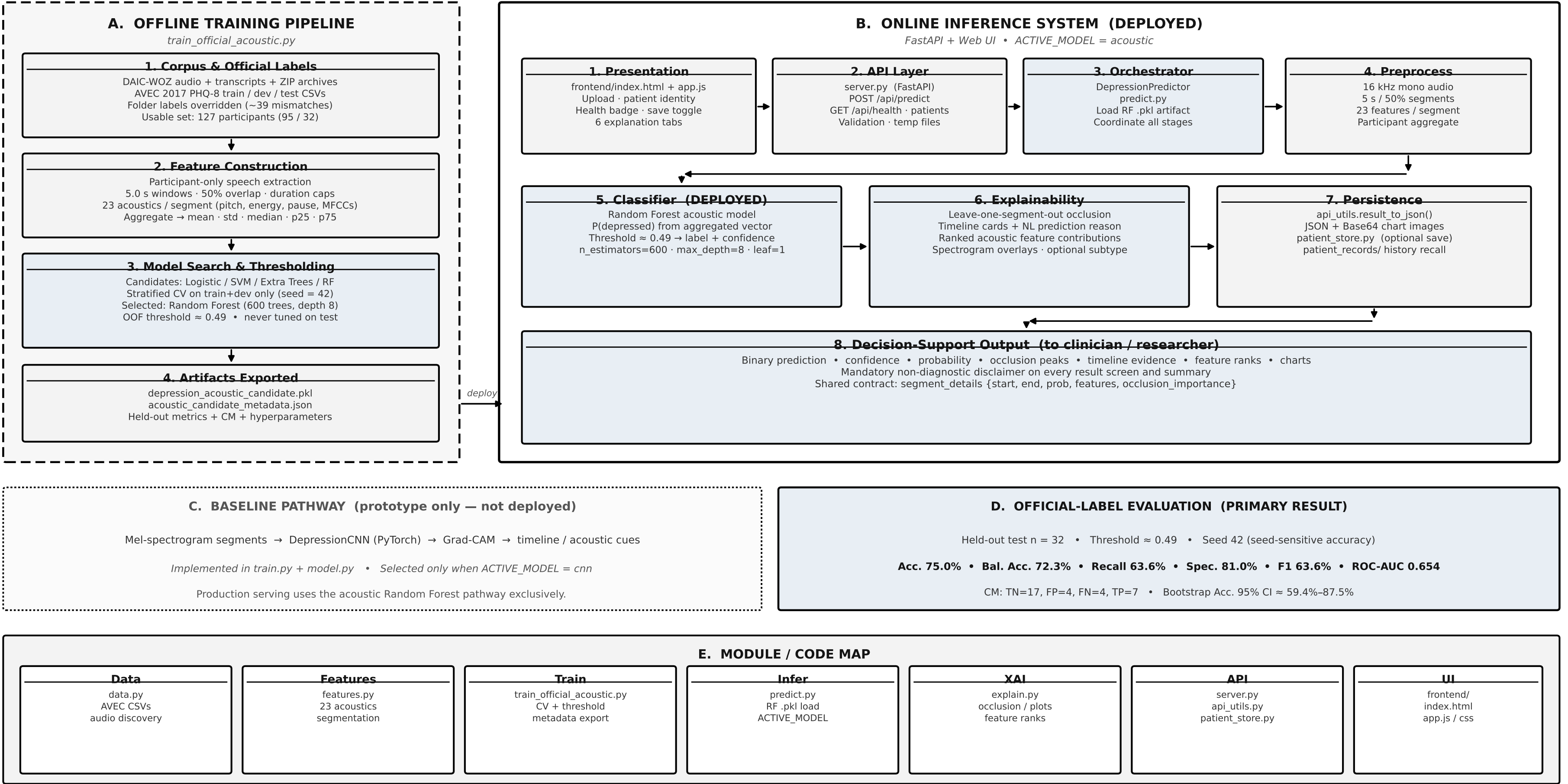


Full Top-Tier System Architecture

Explainable Speech-Based Depression Detection Framework • Deployed pathway: Random Forest acoustic model



Legend



Deployed / key



Supporting

----- Offline training

..... Baseline only



Online deployed system

Design principles: separation of concerns • leakage-aware official-label evaluation • model-faithful occlusion XAI • decision-support framing (not diagnosis)

Previous Extra Trees candidate (not deployed): Acc. 59.4%, recall 63.6%. Accuracy of the deployed RF is seed-sensitive and should be reported with the bootstrap CI.